

ECON 139: Labor Economics

Problem Set 1

Note on Grading: Homework is graded using check +, check or check -. If you answer all questions and generally show you have put thought and effort into the problem set, you will get a check + (10/10). If your answers are sometimes incomplete, or show a clear lack of effort in parts, you will receive a check (7/10). If your answers are often incomplete and show little effort overall, you will get a check - (0/10). The total number of points for each problem set in the course is 10 points. If you turn this in within one day late, the highest score you can get is a check (7/10). No credit is given for assignments that are turned in more than one day late.

Note on AI Use: As discussed in the syllabus, you can use AI tools to help with this problem set. For the first part, however, I highly recommend you complete this on your own without any AI assistance. This type of problem may very well appear on a midterm or final. If you cannot solve it yourself, you will likely struggle on the exam.

For the empirical part, I recommend using AI tools. Not everything in the empirical question comes directly from the lecture. You will need to figure out some parts on your own (as you will often need to do in the future!). AI tools will be useful for this.

1 Part 1: Perfect Competition

Background: You are analyzing the labor market for baristas at a University campus. There are two competing coffee shops: Copa Vida and Tec Cafe. Both shops sell lattes at \$5 each and operate in a perfectly competitive labor market for baristas. Each cafe currently operates with 2 espresso machines ($K = 2$). K is the symbol we generally use for capital, which in this case is espresso machines.

The production function for both cafes follows the form $Y = f(K, L)$, where K is the number of espresso machines and L is the number of baristas. The production data for both cafes with $K = 2$ machines is given below:

Number of Baristas (L)	Copa Vida Lattes/hour	Tec Cafe Lattes/hour
1	10	8
2	18	14
3	25	19
4	31	23
5	36	26
6	40	28
7	43	29
8	45	30

Questions:

- Construct a table that displays the marginal revenue product of labor (MRPL) for each barista at both cafes, given that lattes sell for \$5 each.
- Draw a figure that displays the market-level demand curve (you can draw this from a wage of \$10 to \$50). It might be easiest to first draw the individual-level demand curves and then aggregate them.
 - Hint: To do this, figure out how many workers are hired for each cafe at the following wage levels: \$50, \$40, \$30, \$20, and \$10, and \$10. Then sum the quantities at each of these wage levels to get total market demand. For some values of the wage you may find that the cafe would definitely like to hire the third worker (i.e. $MRPL > W$ for 3rd worker), but would not want to hire the fourth worker (i.e. $MRPL < W$ for 4th worker). In this case, you can assume that the cafe hires 3 workers (i.e. not something like 3.5 workers)
- If the number of workers will to work at a wage of w is equal to $\frac{w}{2}$ (i.e. if the wage is \$10 there are 5 willing workers), what is the equilibrium wage and quantity of baristas hired in this market? (Note: in the previous question you should have found the demand at wages of \$10, \$20, \$30, \$40, and \$50. You can use this information to help you find the equilibrium wage and quantity).
- Now suppose Tec Cafe decides to buy a third espresso machine. This third espresso machine will boost productivity so that Tec Cafes new production function is given below. Given this new production function, will the market wage increase, decrease, or stay the same? Explain why. You do not need to find the new equilibrium wage, just say if it will increase, decrease, or stay the same and explain why. Again you do not need to find the new equilibrium wage, but it will be useful to redraw market demand for wages equal to \$10, \$20, \$30, \$40, and \$50 to help you answer this question.

Number of Baristas (L)	Tec Cafe Lattes/hour with 3 machines
1	12
2	21
3	28
4	34
5	39
6	43
7	46
8	48

- Imagine the Tec cafe has one of their espresso machines break. They decide to buy a new type of espresso machine that automates much of the process of making a latte. This machine boosts productivity (but differently than just adding another machine). The new production function for Tec Cafe is given below. Given this new production function, will the market wage increase, decrease, or stay the same? Explain why. If your answer is different than your answer to the previous question, explain why.

Number of Baristas (L)	Tec Cafe Lattes/hour with New Machine
1	20
2	25
3	29
4	32
5	35
6	37
7	38
8	39

Again for this question it may be useful to draw the new market-level demand curve for wages equal to \$10, \$20, \$30, \$40, and \$50.

2 Part 2: Empirical Exercise

For this question you should submit either (1) an excel file that has executed every step below (i.e. note if you are using Excel, you should download the .xlsx version of the data. While csv files also work in Excel, when you save your file it won't save the formulas or figures you made, instead saving only the data). If you use a different software (Stata, R, Python), you should submit a do file, R script, or Python script that executes every step below. If you use code, no need to include the final histogram as a separate file, we can see if it is correct from your code.

This question requires using two datasets: `cps_aug_25.xlsx` (or `cps_aug_25.csv` if you are using R or Python or `cps_aug_25.dta` if you are using Stata). We are going to merge this dataset with the minimum wage levels in each state (`min_wage.xlsx` or `min_wage.csv` or `min_wage.dta`). Both of these datasets are available in Week 2 modules (note: if you have previously downloaded the cps data, you will need to re-download. The revised data should include a variable called `STATEFIP` which is the state FIPS code that will allow us to merge the two datasets).

- Step 1: Add a column in the CPS dataset that contains the minimum wage in the state where each individual lives. You should not do this manually. You need to figure out how to combine the two datasets together based on the variable `STATEFIP`, which appears in both datasets. This is an extremely common, but often difficult, data task.
 - Adding columns to a dataset is generally referred to as merging or joining datasets. In Stata, you can use the `merge` command. In R, you can use the `dplyr` package's `left_join()` function. In Python (pandas), you can use the `merge()` function.
 - In Excel you can use the `VLOOKUP()` function to do this. If you have not done this before it can be quite tricky! I recommend using AI tools frequently. If something unexpected happens, ask why. Tell the AI tool what type of computer you have, etc. The more specific you are, the better the answer will be. If you don't understand the formula it gives you, ask specifically where you need to enter it and what every component of the formula does.
- Step 2: Generate a new variable called `WAGEHOUR` that is equal to the hourly wage of each individual. You can calculate this by dividing `EARNWEEK2` by `UHRSWORKT` (note if you are using Stata, the variables are not all capitalized, implying `EARNWEEK2` is actually `earnweek2` and `UHRSWORKT` is actually `uhrsworkt`)

3. Step 3: Create a new variable called `WAGE_REL_MIN` which is equal to the individual's hourly wage (`WAGEHOUR`) divided by the minimum wage in their state (`MIN_WAGE`). This variable tells us how many times the minimum wage each individual earns. For example, if someone earns \$15/hour and the minimum wage in their state is \$10/hour, then this variable would be equal to 1.5 (i.e. they earn 1.5 times the minimum wage).
4. Step 4: Restrict the sample to only include individuals who have non-missing values for `EARNWEEK2` (remember missing values are coded as 999999.99) and only include individuals who work exactly 40 hours per week (`UHRSWORKT` is equal to the usual amount of hours the individual works). In Excel, if you highlight all columns and then go to Data → Filter, you can then filter each column to only include certain values. If you do it one at a time, it will often reset your filter (on Mac especially).
5. Step 5: Create a properly formatted histogram of the variable `WAGE_REL_MIN`.